Supabase Backup & Export SOP (Schema, Policies, Views, Functions, and Data)

Purpose

This document provides the complete process for:

- Logging into Supabase CLI
- Linking a project
- Exporting database schema
- Exporting database data
- Creating a complete backup of a Supabase project

Prerequisites

1. Verify Supabase CLI Installation

Open CMD or PowerShell and run:

```
supabase --version
```

Expected output:

```
2.x.x
```

If Supabase CLI is not installed:

```
npm install -g supabase
```

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2. Verify PostgreSQL Client Tools

Check whether pg_dump is available:

```
pg_dump --version
```

Expected output:

```
pg_dump (PostgreSQL) 17.x
```

If not installed:

- Install PostgreSQL
- Add PostgreSQL's bin folder to PATH

Typical location:

```
C:\Program Files\PostgreSQL\17\bin
```

Restart CMD after updating PATH.

Step 1 – Login to Supabase

Run:

```
supabase login
```

A browser window will open.

Generate an access token from:

https://supabase.com/dashboard/account/tokens

Paste the token into the terminal when prompted.

Step 2 – Navigate to Project Directory

Example:

```
cd "C:\Users\vkshu\CRITICAL\Equity Indian Market Framework\tradeos-v6-
complete\tradeos-v6"
```

Step 3 – Find Project Reference

List all accessible projects:

```
supabase projects list
```

Example:

```
NAME            PROJECT REF
tradeos-prod    abcdefghijklmnop
```

Copy the Project Ref value.

Step 4 – Link Local Folder to Supabase Project

Run:

```
supabase link --project-ref abcdefghijklmnop
```

Replace:

```
abcdefghijklmnop
```

with your actual Project Ref.

You will be prompted for the database password.

Retrieve it from:

Supabase Dashboard → Project Settings → Database

Step 5 – Verify Connection

Run:

```
supabase status
```

and

```
supabase projects list
```

Confirm the project is linked successfully.

Step 6 – Create Backup Folder

```
mkdir backups
cd backups
```

All backup files will be stored here.

Export Options

Option A – Export Schema Only (Recommended)

This exports:

- Tables
- Views
- Materialized Views
- Functions
- Triggers
- Indexes
- Constraints
- Sequences
- RLS Policies
- Permissions
- Extensions

No table data is exported.

Command:

```
supabase db dump --linked -f tradeos_schema.sql
```

Output:

```
tradeos_schema.sql
```

Option B – Export Data Only

This exports only table data.

No tables, views, policies, or functions are included.

Command:

```
supabase db dump --linked --data-only -f tradeos_data.sql
```

Output:

```
tradeos_data.sql
```

Option C – Export Complete Database

This exports:

- Full schema
- Policies
- Functions
- Views
- Triggers
- Extensions
- All data

Command:

```
supabase db dump --linked -f tradeos_full_backup.sql
```

Output:

```
tradeos_full_backup.sql
```

Verify Backup Files

Run:

```
dir
```

Expected output:

```
tradeos_schema.sql
tradeos_data.sql
tradeos_full_backup.sql
```

Useful Database Inspection Queries

List All Tables

```sql
SELECT table_name
FROM information_schema.tables
WHERE table_schema='public'
ORDER BY table_name;
```

List All Views

```sql
SELECT table_name
FROM information_schema.views
WHERE table_schema='public';
```

List All RLS Policies

```sql
SELECT *
FROM pg_policies;
```

List All Functions

```sql
SELECT routine_name
FROM information_schema.routines
WHERE routine_schema='public';
```

Recommended Backup Strategy for TradeOS

Before any major release, schema change, migration, or AI framework update:

Run:

```
supabase db dump --linked -f tradeos_schema.sql

supabase db dump --linked --data-only -f tradeos_data.sql
```

```
supabase db dump --linked -f tradeos_full_backup.sql
```

Archive these files with the release version.

Example:

```
Backups
├── TradeOS_v6.0_schema.sql
├── TradeOS_v6.0_data.sql
├── TradeOS_v6.0_full_backup.sql
```

This provides a complete recovery path for both database structure and data.